

# Decision framing, evidence boundaries, tool readiness

Quantitative Methods for Business Analysis. The essentials of every section in one short read: ideas, worked examples and checks.

≈ 17 min read

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Full edition

## 1 · READ

Open a section, read its central idea, then its worked example.

## 2 · APPLY

Try the guided practice and the lab task linked in each section.

## 3 · CHECK

Answer the study check in the writing space without looking back.

## Your study route

### 1.1

#### Quantitative analysis as a decision discipline

8 blocks · Part A

### 1.2

#### From a managerial concern to a bounded question

8 blocks · Part E

### 1.3

#### Testable claims, assumptions and evidence boundaries

11 blocks · Part D

### 1.4

#### Minimum viable readiness in Excel and SPSS

9 blocks · Part B · Part C · Part F

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## 1.1 · Quantitative analysis as a decision discipline

**Learning purpose:** Separate a messy request into problem, decision, question and task, and keep evidence distinct from judgment.

### CENTRAL IDEA

Analysis creates value only when evidence reduces uncertainty for an accountable decision, so the decision comes before the tool.

### IN CLASS

#### [Session slides](#)

The eight-step workflow 13

Missing leading zeros 34

### PRACTISE IT IN THE LAB

- [Part A: What does one row of the order records represent?](#) · 8 min

About 5 min read · PDF pages 4–7 · [Full explanation \(Full Toolkit\)](#)

## Start with the decision, not the software

Managers often send requests that mix a symptom, a preferred solution and a software command, such as "Open SPSS, analyze our support tickets, and stop customer churn." Before opening any file, separate the request into four levels.

| Level                | Meaning                                                  | Example from OmniRetail                                                                       |
|----------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Business problem     | An observed condition needing attention                  | Quarterly cancellations rose 14 percent                                                       |
| Stakeholder decision | A commitment of money, staff or policy owned by a leader | Whether to spend 450,000 dollars a year on 40 weekend support representatives                 |
| Analytic question    | The evidence needed to reduce uncertainty                | Do subscribers with unresolved weekend tickets cancel more often than comparable subscribers? |
| Software task        | A mechanical operation in a tool                         | Filter and summarize records in SPSS or Excel                                                 |

Only the stakeholder decision commits resources. Opening software before the decision is clear is the **tool-first trap**: an accurate calculation can still answer the wrong question. A useful test for any request is: "What specific managerial decision will change depending on the outcome of this calculation?"

The levels sit inside an **evidence-to-decision chain**: condition, stakeholder, decision, uncertainty, evidence question, analysis, bounded interpretation and, last, human judgment. The software task lives inside "analysis," late in the chain.

## Worked example: repairing a tool-first request

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OmniRetail is a simulated subscription retailer with 2.4 million active accounts. After a 14 percent rise in quarterly cancellations, the Vice President of Customer Operations proposed 40 weekend representatives at 450,000 dollars a year, citing a Monday backlog of 4,200 unresolved tickets. A manager then wrote: "Run SPSS on every ticket and stop churn."

The analyst finds four defects: a tool named before the problem, an impossible guarantee ("stop churn"), no decision owner, and no population, comparison or time horizon. The repaired question is: "Among active monthly subscribers, is an unresolved weekend support ticket associated with a meaningful increase in 60-day cancellation rates compared to weekday tickets?" It names the decision it serves, a population, a comparison and a fixed 60-day window. Without that window, the definition of churn drifts and results cannot be reconciled across tools.

Notice what is still unknown: whether each support-log row is one ticket or one customer. That is checked at the next step of the workflow.

## The eight-step workflow

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The course uses eight steps as the spine of every analysis:

- **Step 1, Frame:** clarify the problem, decision and what will be measured.
- **Step 2, Inspect:** audit provenance (where data came from), grain (what one row represents), missing values and baseline quality.
- **Steps 3 and 4, Choose and Assume:** select a valid method and record the conditions it needs.
- **Step 5, Analyze:** run transparent, reproducible calculations.
- **Steps 6 and 7, Quantify Uncertainty and Recommend:** state how uncertain the result is and derive a bounded action.
- **Step 8, Communicate:** turn findings into a defensible recommendation.

Each step is a quality gate. If Step 1 misframes the decision, Step 2 inspects the wrong population. Session 01 practices Frame and Inspect, plus tool readiness; methods arrive in later sessions.

## Evidence informs; people decide

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**Quantitative evidence** reduces uncertainty. **Managerial judgment** belongs to the accountable executive, who weighs evidence against budget, feasibility, contracts, legal requirements and ethics. The slides illustrate this: even if customers with weekend ticket delays showed a 6 percent higher cancellation rate, leaders would still compare 450,000 dollars a year with alternatives such as self-service knowledge bases or chatbot triage. The guided demonstration gives a sentence to complete for any request: "The evidence can show , *but the decision maker must still judge* ."

### COMMON MISTAKE Common misconceptions

- "Analysis starts when the file opens." Opening a workbook is a software task, not an analysis.
- "More data is more credible." Irrelevant or poorly documented records add noise and risk.
- "A significant finding is a mandate." Statistical significance does not equal mandatory action; leaders may reasonably decline a sound recommendation.

### LIMITS AND CONDITIONS Limits

Some requests mix several decisions; write one frame per decision. If nobody can name a decision, ask what action or allocation could change. When evidence is inaccessible, prohibited, irrelevant or likely to mislead, pausing is the professional response. Records can show when demand occurred, but they cannot decide budget tolerance or staffing feasibility.

## Connect to Part A

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Part A of the S01P practice applies Step 2, Inspect, to 120 MetroFresh order records. On the **Data Audit** sheet, write what one row represents, which field identifies it and whether a repeated customer\_id is an error or expected (one customer can place several orders). Check the rule that items\_unfulfilled never exceeds items\_ordered, and give each of five fields a measurement level, reason and treatment in B15:D19. In SPSS, run the GET DATA block and confirm 120 cases. Never edit the Data sheet.

## Study actions

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**Reflect:** Do your work habits begin with a preferred tool, an available file or a stakeholder decision? What would change if you always started with the decision?

**Recall without looking back:** Classify these four requests: "Quarterly cancellations have risen"; "Should the service vice president add weekend coverage next month?"; "What approved evidence would reduce uncertainty about whether weekend demand warrants a coverage change?"; "Open the workbook and run the provided step."

**Lab connection:** Bring one sentence that says what one row of the MetroFresh records represents.

## 1.2 · From a managerial concern to a bounded question

**Learning purpose:** Turn a vague concern into a question with a stated owner, scope, time horizon, outcome and comparison.

### CENTRAL IDEA

A question must be broad enough to matter and narrow enough to answer honestly, and its boundaries must be written down before any data are touched.

### IN CLASS

#### [Session slides](#)

Evidence and managerial judgment 14

Causal claims and observational data 22

### PRACTISE IT IN THE LAB

- [Part E: Can we use these records as evidence, or should we pause?](#) · 10 min

About 4 min read · PDF pages 8–11 · [Full explanation \(Full Toolkit\)](#)

## The anatomy of a bounded question

A **bounded analytic question** states enough context, scope and time to limit what the answer represents. The slides group its parts into three tiers.

| Tier             | Components                                                   | Purpose                                                       |
|------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Decision Context | Stakeholder, pending decision, minimum actionable difference | Anchors the question to a real commitment                     |
| Evidence Scope   | Population, explicit exclusions, time horizon                | States whose experience and which period the answer describes |
| Measurement Plan | Observable outcome, reference point, verified data source    | States what will be observed and against what                 |

The **stakeholder** is the role accountable for using the evidence. The **decision** need not be yes or no; it may be choosing, prioritizing, allocating, revising or pausing. The **reference point** is a comparison, such as a prior period or another group; without one, you cannot tell whether an observed rate is elevated. In this



session you name the outcome and reference point without building detailed measures; metrics, variables and grain come next session.

## Worked example: the OmniRetail frame

| Component                     | OmniRetail entry                                                       |
|-------------------------------|------------------------------------------------------------------------|
| Stakeholder                   | Vice President of Customer Operations                                  |
| Decision                      | Allocate 450,000 dollars a year for weekend support                    |
| Minimum actionable difference | At least five percentage points higher cancellation, agreed in advance |
| Population                    | Active United States monthly subscribers                               |
| Exclusion                     | Enterprise accounts                                                    |
| Time horizon                  | Trailing twelve months, with a 60-day outcome window                   |
| Outcome and reference         | 60-day cancellation rate, compared with resolved weekday tickets       |

A **percentage point** is the arithmetic difference between two percentages: 10 percent to 15 percent is five percentage points. Compare the vague question "Do weekend tickets hurt retention?" with the bounded one: "Among active United States monthly consumer subscribers over the trailing twelve months, is an unresolved weekend support ticket associated with a higher 60-day cancellation rate compared to resolved weekday tickets?" The bounded version adds a population, an exclusion, a horizon, a window and a comparison.

A second example from the session shows that the same concern can support different frames. For "Improve customer retention," a customer success director deciding whether to commission a limited onboarding pilot might ask what approved account records for United States monthly subscribers enrolled from January through March show about documented continuation 60 days after enrollment, relative to an agreed prior-period reference. Annual plans, other countries and behavior after 60 days are outside the frame. A marketing leader deciding whether to revise acquisition messaging would need a different question, because the decision differs.

## A short framing workflow

1. Ask what decision could change, who owns it and when it must be made.
2. Translate the decision into the uncertainty the evidence should reduce.
3. State the included and excluded context and the time horizon.

4. **Replace vague words** such as better, successful, recent and customers with bounded language.
5. **Name the observable outcome and reference point**, without building a detailed metric yet.
6. **Check** relevance, testability, boundaries, feasible evidence and neutral wording.

If two plausible readings would lead to different decisions, pause and consult the stakeholder. If you must set a boundary yourself, label it as an assumption.

## Keep scope under control

The session names three scoping mistakes.

- **Uncontrolled scope drift:** stakeholders add questions, cohorts and metrics without moving deadlines. In the slide example, adding shipping carrier delays to the OmniRetail study would delay the staffing decision by eight weeks.
- **Premature metric coding:** formulas are written before teams agree what terms such as "active subscriber" mean, producing conflicting numbers.
- **Concealed limitations:** gaps or exclusions are hidden, so leaders assume the findings apply everywhere.

The remedy for drift is triage, not refusal: sort each addition into **Deliver Now**, **Phase Later** or **Separate Charter** and show the timeline impact. An addition needs its own charter when it involves a different data source, owner or budget. Log every excluded observation with a rule, a reason and an exact count, and put limitations in the summary rather than an appendix.

### **COMMON MISTAKE** Common misconceptions

- "Available data should set the question." Data affect feasibility; the decision sets relevance.
- "Broader is more strategic." Breadth can merge incompatible decisions.
- "Say yes to every addition." Accepting everything stalls the core decision.

### **LIMITS AND CONDITIONS** Limits

Every exclusion narrows what the finding covers. Excluding enterprise accounts makes the OmniRetail comparison cleaner and also means it cannot justify enterprise support policy. Bounding cannot create missing evidence; if the preferred source is unavailable, document it, negotiate an approved proxy or revise the scope.

## Connect to Part E

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Part E asks for a 150 to 200 word evidence-sufficiency note in **Synthesis!A5**. The Vice President's email asks whether the approved order records can be used as evidence, not whether to change service coverage. Staying inside that scope is the skill: give a clear use-or-pause recommendation, at least three numbers you can trace to a cell, what the records can describe, what they cannot establish and the main limitation.

## Study actions

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**Reflect:** A stakeholder asks you to add four new dimensions a week before the deadline. What would you deliver now, and what would you phase?

**Recall without looking back:** Name the three tiers of a bounded question and one component of each.

**Lab connection:** Bring your use-or-pause recommendation in one sentence.

## 1.3 · Testable claims, assumptions and evidence boundaries

**Learning purpose:** Write claims that evidence could challenge, and match their strength to what the evidence can support.

### CENTRAL IDEA

Persuasive wording cannot strengthen weak evidence; a claim is only as strong as the scope and design behind it.

### IN CLASS

[Session slides](#)

Assumptions and limitations 21

MVR quality checklist 32

### PRACTISE IT IN THE LAB

- [Part D: Which records need a second look?](#) · 8 min

About 5 min read · PDF pages 12–15 · [Full explanation \(Full Toolkit\)](#)

## Make the claim challengeable

A **provisional claim** states what the evidence might show, written before looking. It is **testable**, or **falsifiable**, when an observable result could count against it. "Customer service quality matters" is not testable: no observation could contradict it. Claim strength cannot exceed the relevance, provenance, scope, timing or design of the evidence.

A **claim audit** checks scope, comparison, direction, time horizon and decision relevance, then provenance and design. Flag causal words (caused, produced, led to, resulted in) and certainty words (will, every, eliminate). Before querying data, complete: "My claim would be challenged if..."

## Worked example: OmniRetail's claim and threshold

The claim: "Active monthly subscribers experiencing unresolved weekend support tickets exhibit a 60-day cancellation rate at least five percentage points higher than those with resolved weekday tickets." The threshold of five points is agreed with leadership before any query. The slides illustrate the disconfirmation

rule: an observed difference of only 1.2 percentage points means weekend delays do not justify the 450,000-dollar budget. Disconfirming a sponsor's hypothesis is a useful result, not a failure.

Written as a rule, with  $\Delta$  the observed difference and  $\delta$  the agreed threshold:

$$\Delta = p_{\text{weekend, unresolved}} - p_{\text{weekday, resolved}}, \quad \text{supports the commitment only if } \Delta \geq \delta = 5$$

## Significant is not the same as worth acting on

**Statistical significance** means a difference is unlikely to be sampling noise. With millions of records, tiny differences qualify. In the session's illustration, across 2.4 million records, a 0.3-percentage-point difference is significant at  $p < 0.001$ , yet the slides value it at about 24,000 dollars of saved annual revenue against a 450,000-dollar cost. **Practical significance** asks whether a difference is large enough to justify action, which is why the threshold is agreed before results are seen.

 LIMITS AND CONDITIONS

Assumptions to validate, limitations to disclose

| Type       | Meaning                                          | OmniRetail example                                           | Action                                                      |
|------------|--------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|
| Assumption | A condition accepted for now that can be checked | Ticket timestamps record when customers experienced friction | Validate against server error logs                          |
| Limitation | A boundary that the data or design cannot remove | Only United States consumer subscribers are studied          | Disclose: findings cannot justify enterprise support policy |

The podcast summarizes the difference: you audit assumptions in a log; you transparently disclose limitations. A limitation narrows a valid conclusion without making it worthless.

## Worked example: revising a causal claim

The claim "The new onboarding program caused customers to renew" rests on observed participation records, which show who took part and who renewed but not why. The bounded revision: "Within the approved population and period, documented renewal was more common among customers recorded as participants than among those recorded as nonparticipants; these records alone do not establish why." Assume statuses were recorded consistently and check the documentation; disclose that one region and period may not represent others. The claim would be challenged if the scoped records showed no such pattern or the opposite.



## GO DEEPER

### Why observational patterns mislead

Most business records are **observational**: they show associations and sequences, not causes. The slides' example: subscribers who joined with a discount code show 35 percent higher cancellation. The likely driver is **price sensitivity**, which leads customers both to seek coupons and to cancel when full billing begins. It is a **confounder**, a third variable affecting both the supposed cause and the outcome. Removing discounts would not create loyal customers.

Use disciplined wording such as "was associated with," "co-occurred with" or "differed between groups," and list at least two alternative explanations. Replacing "caused" with "drove" or "impacted" still implies cause.

## A short claim workflow

1. Write the provisional claim from the bounded question.
2. Agree and date a practical threshold with the decision owner.
3. Write the disconfirming observation.
4. Audit scope, comparison, direction, horizon and relevance; flag causal and certainty words.
5. Log assumptions with validation steps and limitations with boundary statements.
6. Sort the claim as supportable, needs qualification or unsupported, and revise.



## COMMON MISTAKE

### Common misconceptions

- "Careful wording fixes weak evidence." It bounds a claim but cannot repair unsuitable evidence.
- "Listing limitations weakens a recommendation." It strengthens credibility by preventing overuse.
- "A significant regression proves cause." In observational data it shows association after accounting for the included variables.



## LIMITS AND CONDITIONS

### Limits

Causal language needs a supporting design, such as a randomized experiment. When records show only an association, the session proposes a **controlled operational pilot**: for example, a 90-day weekend support pilot in three representative regions, with the other regions as a control group,

tracking First-Contact Resolution, repeat contacts and 60-day cancellation. Its window, sample size and method are fixed before launch. When missing or unsuitable evidence would make any result misleading, recommend no immediate analysis.

## Connect to Part D

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Part D asks which records need a second look. On the **Quality Log** sheet, log each blank field, the largest delivery time and the result of the items rule. Keep every value exactly as recorded: the workbook's rule is not to replace blanks with zero, delete unusual values or alter source records. For each row, separate what you can check later (an assumption, such as whether a delivery time was recorded correctly) from what you must disclose (a limitation, such as the records covering only four weeks). Write Part E in associational language.

## Study actions

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**Reflect:** Your sponsor's preferred hypothesis falls short of the agreed threshold. How would you report it, and what would you say about next steps?

**Recall without looking back:** Why is timestamp accuracy an assumption, while consumer-only scope is a limitation?

**Lab connection:** Bring one flagged record and the question that remains about it.

## 1.4 · Minimum viable readiness in Excel and SPSS

**Learning purpose:** Produce saved, reopened, traceable artifacts in both tools and decide honestly whether work is ready to proceed.

### CENTRAL IDEA

Readiness is proven by artifacts another analyst can reopen and repeat, not by a launched application or a screenshot.

### IN CLASS

#### [Session slides](#)

Excel and SPSS roles 29

Spreadsheet modeling standards 30

Cross-tool reproducibility 49

SPSS workspace architecture 31

Readiness trace 33

### PRACTISE IT IN THE LAB

- [Part B: What do delivery time, order value and spoilage look like?](#) · 12 min
- [Part C: Do Excel and SPSS give the same answer?](#) · 10 min · SPSS Descriptive Statistics
- [Part F: Where does each number in the note come from?](#) · 6 min

About 5 min read · PDF pages 16–19 · [Full explanation \(Full Toolkit\)](#)

## What readiness requires

Minimum Viable Readiness (MVR) has three pillars in the slides:

- **Source protection:** keep the raw file untouched and work in a named copy.
- **Dynamic execution:** an Excel input change updates linked outputs, and saved SPSS syntax runs cleanly.
- **Evidence verification:** close, reopen and confirm that results still appear without errors.

In Excel, keep three layers separate: labeled **inputs**, **logic** formulas that refer only to those inputs, and an **output** summary. A number typed inside a formula is a buried constant; it hides its meaning and is missed when assumptions change. In SPSS, three separate windows hold three separate files: the **Data Editor** (.sav, with Data View and Variable View), the **Syntax Editor** (.sps, commands) and the **Output Viewer** (.spv, tables, logs and warnings). Saving one does not save the others. In dialogs, click **Paste** rather than OK, so the command is saved as syntax.



## Worked example: the two readiness traces

In the Excel trace, open the working copy, find the designated input, make a reversible change, watch the linked output respond, restore the value and save. The slides change an hourly support wage from 20 to 25 dollars and back. In the SPSS trace, open the data and syntax separately, run only the named block, read the log at the top of the Output Viewer, save each file, close everything and reopen from the locations in your readiness log. The log records source, file name, action, observed result, saved location and any unresolved issue.

If SPSS syntax fails, check that every command ends with a period and that variable names match Variable View. If an identifier such as 00458921 imports as 458921, a display format will not fix it; set the column to Text in Power Query, or define a String variable in SPSS.

## Worked example: MetroFresh parity

The practice workbook holds 120 MetroFresh orders from August 1 to 28, 2026, placed by 76 distinct customers. SPSS reads the same **Data** sheet through the supplied syntax. The practice guide gives these check values:

| Variable          | Total N | Valid n | Missing | Mean   | Sample SD |
|-------------------|---------|---------|---------|--------|-----------|
| delivery_minutes  | 120     | 118     | 2       | 43.67  | 14.17     |
| order_value (USD) | 120     | 119     | 1       | 142.07 | 57.27     |
| spoilage_units    | 120     | 120     | 0       | 0.65   | 1.29      |

The row check shows no record was lost: valid plus missing equals 120 for every variable. **Cross-tool parity** compares fifteen numbers (valid n, mean, minimum, maximum and sample SD for three variables). Each Excel and SPSS pair must agree within 0.01:

$$| \text{Excel} - \text{SPSS} | \leq 0.01 \Rightarrow \text{PASS}$$

The divisor matters. Excel's STDEV.S divides by n minus 1, like SPSS, and gives 14.17 for delivery time. STDEV.P divides by n and gives 14.11. The gap of 0.06 fails the tolerance, so that row would show CHECK.

## A short readiness workflow

1. Put the workbook and syntax file in one folder and save a working copy under your own name.
2. Run GET DATA and confirm 120 cases before reading any table.

3. Build Excel results with formulas that point at the table; never type results.
4. Reconcile counts first, then compare summaries across tools.
5. Investigate every CHECK through rules (blanks, filters, divisors), not by retyping values.
6. Save, close, reopen and log each artifact.

## Deciding whether to proceed

The session ends with four **readiness classifications**:

| Classification         | When to use it                                                                          |
|------------------------|-----------------------------------------------------------------------------------------|
| Ready to analyze       | The question is bounded, evidence is feasible and permitted, and the workflow functions |
| Clarification required | Intent, decision use, scope or acceptable claim strength is still ambiguous             |
| Evidence blocked       | Needed evidence is missing, prohibited or unsuitable for the required claim             |
| Tool blocked           | The workflow cannot produce persistent, inspectable artifacts                           |

In the session's integrated case, a regional operations director wants existing files to "prove" that flexible scheduling will improve service. If locations, period or outcome are unspecified, choose clarification required; if causal proof is demanded but the records cannot provide it, choose evidence blocked. A documented pause is valid professional work.

### COMMON MISTAKE Common misconceptions

- "A screenshot proves it works." It cannot show dynamic formulas, repeatable syntax or a file that reopens.
- "The file that opens is the right file." If instructions name version 1 but the file shows version 2, stop, record both and ask.
- "PASS proves the data are right." A wrong value gives the same wrong mean in both tools.

### LIMITS AND CONDITIONS Limits

Parity confirms consistent processing, not accurate records or fitness for the decision. A clean import confirms access, not provenance. The MetroFresh records describe four weeks of orders; they cannot show why any delivery took longer, and the 142-minute delivery time is a value to check, not delete.

Python and R appendices are optional reference checks only.

## Connect to Parts B, C and F

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In Part B, fill **Descriptive Statistics** rows 6 to 8 with formulas; D6 should show 118, E6 should show 2 and K6 should show 14.17. In Part C, link column C of **Excel-SPSS Parity** to your Excel results by formula, type the SPSS values into column D and reach fifteen PASS results. In Part F, give every number in your note its own row in the **Memo Evidence Map**, with a workbook cell and an SPSS source, and type Yes only after reopening both files.

## Study actions

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**Reflect:** Which step in your own file workflow is least reliable, and what recovery action would you record in your readiness log?

**Recall without looking back:** Why does saving an SPSS dataset not save your output? What does a PASS on the parity sheet prove, and what does it not prove?

**Lab connection:** Bring the valid N and mean for delivery time, and the number of parity checks that passed.